CSAI 490- PRD:

VOYO - Your AI Guide to Egypt

Authors: Youssef Hesham, Yaseen El-Beltagy, Youssef Amr, & Mahmoud Nazmy

I. Introduction: The Traveler's Dilemma in a Land of Wonders

Egypt: a land of ancient wonders, vibrant culture, and timeless stories. For countless travelers, it's the destination of a lifetime. Yet, for too many, this dream is shattered by a harsh reality of tourist traps, persistent scams, and the daunting challenge of navigating a complex country alone. The difference between an unforgettable journey and a regrettable trip often comes down to one thing: **trusted, local knowledge.**

This is the critical gap WANDRD was born to fill. Foreign travelers, lacking reliable guidance, are often left vulnerable, funneled into bland experiences that barely scratch the surface of what Egypt truly offers. This information gap prevents them from discovering authentic cultural gems and navigating the country with the confidence they deserve.

WANDRD is not just another travel app; it is a personalized, AI-powered guardian and guide for the modern traveler. By leveraging a custom agentic AI framework grounded in a curated, expert-vetted database, WANDRD provides a shield against negative experiences and a compass pointing towards authentic, memorable ones. It's a single, intelligent platform that acts as a knowledgeable local guide, available 24/7, right in your pocket.

II. Vision: Redefining Global Travel, Starting with Egypt

Our vision is to empower every traveler to explore the world with the confidence and insight of a seasoned local expert. We aim to replace uncertainty with discovery and anxiety with adventure.

While our initial launch focuses on providing an unparalleled, safe, and authentic experience for travelers in **Egypt**, our ultimate ambition is to scale WANDRD into a **global travel companion**. We will build a framework that can be adapted to any country, bringing the same level of dynamic personalization, safety, and discovery of "hidden gems" to travelers across the world.

III. Project Objectives

For our Senior Project, we will design and build a functional prototype of the WANDRD mobile application with the following key objectives:

1. **Deliver a robust AI Chatbot ("Cleo")** that provides instant, reliable, and personalized travel advice.
2. **Develop an Interactive Map Explorer** as the core discovery tool for both famous landmarks and hidden gems.
3. **Implement an AI-powered Itinerary Planner** that generates logical, efficient, and customized travel plans.
4. **Construct a curated backend database** that serves as the "ground truth" for the AI, prioritizing authenticity and safety.
5. **Demonstrate a seamless User Flow** from onboarding and discovery to planning and real-time

assistance.

IV. Target Audience

- **Solo Adventurers (Ages 20-35):** Tech-savvy, budget-conscious travelers seeking unique experiences and safe ways to explore.
- **Families (Ages 30-55):** Seeking safe, educational, and engaging activities with well-structured, stress-free itineraries.
- **Couples & Honeymooners (Ages 25-45):** Looking for a mix of romantic, cultural, and relaxing experiences.
- **History & Culture Enthusiasts (All Ages):** Seeking deep, accurate information and access to significant historical sites.

V. Core Features & User Flow

5.1. Primary Features (MVP)

Feature	Description	Key User Stories
AI Chatbot Travel Guide ("Cleo")	An AI-powered chatbot that acts as a 24/7 travel expert, providing instant answers and personalized recommendations grounded in a secure database.	- Ask questions in natural language. - Get recommendations tailored to my profile. - Receive practical info (hours, prices, safety tips). - Learn deep historical and cultural context.
Interactive Map Explorer	A visually rich, filterable map for discovering points of interest, viewing curated thematic layers, and exploring pre-made routes.	- See major landmarks on a map. - Filter locations by interests ("Temples", "Diving"). - Tap a location for details and add to my itinerary. - View safe, pre-vetted routes on the map.
Personalized Itinerary Planner	An intelligent tool that generates custom, editable, day-by-day travel itineraries based on user preferences and sophisticated AI logic.	- Input travel dates, interests, and pace to get a plan. - Receive a logical, efficient daily schedule. - Edit the itinerary with a

		drag-and-drop interface. - Save the plan for offline access.

5.2. Secondary Features (Post-MVP)

Feature	Description
Egyptian Food Guide	A dedicated section for discovering authentic Egyptian dishes, from street food to fine dining.
Connect Solo Travelers	An opt-in feature for solo travelers to join small, app-organized group activities safely.
Flight Scanner	An integrated tool to search for flights, helping with initial travel planning.

5.3. User Flow

1. **Onboarding:** A new user signs up and completes a brief questionnaire to establish their travel profile (interests, budget, style).
2. **Discovery (Explore Page):** The user lands on the homepage featuring the Interactive Map. They explore regions and filter by interest. Below the map is a carousel of pre-made itineraries, followed by navigation buttons.
3. **Interaction (Chat):** The user asks Cleo, "I have 3 days in Luxor and love photography. What should I see?"
4. **Planning (Itinerary Generation):** Based on the chat or planner tool, the AI generates a detailed 3-day Luxor itinerary, optimized for logistics and user preferences.
5. **Refinement:** The user easily edits the plan, swapping a temple visit for a hot air balloon ride.
6. **Travel:** The user accesses their saved itinerary offline, uses the map for navigation, and asks Cleo for real-time advice.

VI. Technical Architecture & Stack

- **Frontend (Mobile App): Flutter**
  - **Why:** A modern, cross-platform framework that allows for a single codebase for both iOS and Android. Its high-performance rendering engine and extensive widget library are ideal for building a beautiful and responsive UI, especially for the interactive map and itinerary planner.
- **Backend: Python with FastAPI**
  - **Why:** FastAPI's high performance and native support for asynchronous operations are perfect

for handling real-time requests from the app. Its tight integration with Python's data science and AI/ML ecosystem makes it the ideal choice for our backend.

- **Database: Supabase (PostgreSQL)**
  - **Why:** An open-source Firebase alternative. It provides a powerful SQL database with geospatial support (PostGIS), authentication, and instant APIs out of the box. Ideal for a university project that values open-source technology and relational data structures.
- **AI Integration: Open-Source LLMs (e.g., Llama 3, Mistral, Groq)**
  - **Why:** Provides powerful reasoning capabilities while allowing for fine-tuning and local deployment. This gives us full control over the AI's behavior, ensures cost-effectiveness for a university project, and allows us to build a highly specialized agentic system.
- **External APIs & Services:**
  - **Mapping:** Mapbox or Google Maps API for interactive map tiles, geocoding, and routing.
  - **Weather:** A reliable Weather API (e.g., OpenWeatherMap) to provide real-time weather data for the check_weather tool.
  - **Flights:** Amadeus API for the post-MVP flight scanner feature, providing access to comprehensive flight data.

VII. Core Innovations: The WANDRD Engine

7.1. Agentic AI Framework

We will implement a custom agentic system where our open-source LLM acts as a reasoning engine, using a suite of specialized tools to perform complex tasks and ensure reliability.

- **AI Chatbot ("Cleo") Agent:**
  - **System Prompt:** *"You are Cleo, an expert and friendly guide to Egypt. Your primary goal is to provide safe, authentic, and personalized advice. You must prioritize hidden gems over tourist traps. You must cross-reference information with the internal database to ensure accuracy and avoid scams. You have access to the following tools: get_poi_details, check_weather, find_in_database."*
  - **Tool Use Example:** User asks, "What's a good place for dinner near the Cairo museum that isn't too touristy?" The agent executes the tool find_in_database(collection='restaurants', filters={'area': 'Tahrir Square', 'tourist_score_lte': 3, 'authenticity_score_gte': 4}) and uses the results to formulate a safe, authentic recommendation.
- **Itinerary Planner Agent:**
  - **System Prompt:** *"You are an expert travel logistics planner. Your task is to build a logical, efficient, and enjoyable itinerary. The output must be a valid JSON object. You must use tools to verify travel times, opening hours, and location proximity to create a realistic schedule. Available tools: get_travel_time, get_poi_operating_hours, get_poi_coordinates."*
  - **Tool Use Example:** When planning a day, the agent calls get_travel_time(origin='PointA', destination='PointB') and get_poi_operating_hours(poi_id='PointB') to ensure the schedule is both logical and feasible, preventing wasted time and frustration.

7.2. Database Structure (Supabase/PostgreSQL)

Our database is the "ground truth" that powers the AI's reliability.

- **users Table:** Stores user profiles and travel preferences (JSONB).
- **points_of_interest Table:** A detailed catalogue of sites, restaurants, and activities, each with custom-rated scam_risk and authenticity_score fields, plus geospatial data (PostGIS point).
- **itineraries Table:** Stores user-generated and pre-made itineraries.

Database Schema Overview

1. Purpose of the POI Database

Our Points of Interest (POI) database serves as the **core content system** for the AI-powered travel experience. It is the engine that enables:

- **Accurate Map Exploration:** Fast, coordinate-based indexing.
- **Real-time Planning:** Logic-driven itinerary generation.
- **Context-Rich AI:** Providing the "ground truth" for the AI chat agent.
- **Data Enrichment:** Supporting an ongoing automated pipeline.

> **Architecture Goal:** The database is built to be **modular, scalable, and AI-friendly** — structured enough for reliable queries, yet flexible enough for rapid expansion through automated tooling.

---

2. Design Principles

To ensure long-term viability, the schema adheres to the following principles:

Design Principle	Strategic Result
Single source of truth	Reliable grounding for AI responses; eliminates hallucinations.

Normalized structure	Efficient updates, analytics, and data integrity.
Multilingual by design	Full native support for **Arabic & English**.
Optimized for search	Fast geospatial (PostGIS) and text-based queries.
AI-ready storage	Structured specifically to support automatic summarization and tagging.

3. Core Data Model Summary

The data model centers on **two main entities** (Regions and Sites), supported by satellite tables to handle dynamic and rich content. This allows the product to grow from a basic dataset into a rich, intelligent knowledge graph.

Layer	Entity	Functional Purpose
Primary	regions	Major destinations (e.g., Cairo, Luxor).
Primary	sites	Individual attractions/POIs within regions.
Support	site_hours	Detailed opening/closing times.

Support	site_content	Long-form descriptions for AI narrative generation.
Support	site_tags	User-friendly classification & search filtering.
Support	site_categories	Consistent category taxonomy.
External	external_ids	Mapping to OSM, Wikidata, OpenTripMap.
Media	site_media	Images with proper sourcing/licensing.

4. Table Purpose Breakdown

A. Regions Table

- **Function:** Stores top-level travel destinations.
- **Key Data:** Multilingual naming conventions and central geolocation.
- **Usage:** Filters map content and drives high-level trip planning.

B. Sites Table

- **Function:** The central POI record.
- **Key Data:** Geography (lat/lon), Regional linkage, Visitor metadata (price, duration), Accessibility attributes.
- **Usage:** The backbone of **map queries** and **itinerary ranking**.

C. Site Hours Table

- **Function:** Handles temporal logic.
- **Key Data:** Per-day schedules, seasonal closures.
- **Usage:** Answer complex queries like *“Is this open Tuesday at 5 PM?”*

D. Content Table

- **Function:** Repository for authoritative text in English and Arabic.
- **Key Data:** Source-aware descriptions.
- **Usage:** Enables AI summarization, cultural storytelling, and **vector search** for chatbot grounding.

E. Tags & Categories

- **Categories:** Structured taxonomy (e.g., Historic, Cultural, Museum).
- **Tags:** Flexible attributes (e.g., Photography, Shopping, Family-friendly).
- **Usage:** Powers search bars, filter controls, and personalization algorithms.

F. External IDs

- **Function:** Links internal POIs to external datasets (OpenStreetMap, Wikidata).
- **Usage:** Enables **automated data ingestion** and enrichment from official sources without manual entry.

G. Media

- **Function:** Stores image assets.
- **Usage:** Manages metadata for licensing compliance and visual content ordering.

---

5. Why This Schema Works

The structure has been validated against real-world tourism data needs and future functionality.

User-Facing Feature	Database Enabler
Interactive Map Explorer	Indexed region & geolocation fields.
Itinerary Planner	Granular site hours, visit duration, and accessibility

	data.
AI Travel Assistant	Clean structured data + rich textual content tables.
Automated Pipeline	external_ids table + enrichment-friendly field design.

---

7. Technical Benefits

- **Stack Compatibility:** Works seamlessly on **Supabase (PostgreSQL)**.
- **Geospatial:** Supports geospatial indexing (**PostGIS**-optional).
- **Search:** Enables full-text multilingual search.
- **Performance:** Durable against dataset growth (capable of handling thousands of POIs).
- **Ingestion:** Structured specifically for batch ingestion + AI enrichment.

VIII. Competitive Advantage

Competitor Category	Key Players	Critical Weakness / Market Gap	How WANDRD Wins
The Aggregator Giants	TripAdvisor, Google Travel, Expedia	**The "Sea of Noise."** Overwhelming, untrustworthy, and generic recommendations based on popularity, not curated quality. Plagued by fake reviews and tourist traps.	**Grounded Curation & Trust.** Our AI is grounded in an expert-vetted database with custom scam_risk and authenticity_score fields. We replace noise with a clear signal of quality and safety.

The Itinerary Planners	Wanderlog, TripIt, Lambus	**Logistics, Not Discovery.** These are passive tools for organizing a trip *after* you've done the hard research. They don't help you discover unique, personalized experiences.	**Intelligent Discovery & Planning.** Our agentic AI is a research partner. It actively discovers unique places tailored to you and then uses its tools (checking hours, travel times) to build a truly logical and feasible plan.
The AI Chatbots	GuideGeek, Layla, Roam Around	**Ungrounded & Unreliable AI.** Most are thin wrappers around general-purpose LLMs, making them prone to "hallucinating" facts, recommending closed venues, and repeating generic top-10 search results.	**Reliable, Agentic System.** Our AI is not just a chatbot; it's an agent. It uses our curated database as its "brain," ensuring every recommendation is factually accurate, safe, and aligned with our core mission. It provides **reliable answers, not guesses.**

Our Unique Selling Proposition (USP): WANDRD stands out by combining the **reliability of an expert-curated database** with the power of a **custom-built, agentic AI**. This allows us to offer a level of dynamic personalization, focus on safety, and discovery of authentic "hidden gems" that no current competitor can match.

IX Conclusion

The WANDRD project represents more than just the development of a mobile application; it is an ambitious endeavor to solve a real-world problem for millions of travelers. By integrating cutting-edge agentic AI systems with a foundation of carefully curated, reliable data, we are not merely building a tool but a trusted companion. This project will demonstrate a mastery of modern software engineering principles, advanced AI implementation, and a user-centric design philosophy. We are confident that WANDRD will not only meet the requirements for our senior project but will also lay the groundwork for a commercially viable product that has the potential to enhance and secure the travel experiences of people all over the world.